Physiological biomarkers of stream fish from the Atlantic Forest within a Conservation Unit and its surroundings

Subtitle 1 Subtitle 2

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The Atlantic Forest is one of the most degraded biogeographical provinces of Brazil, and the creation of Conservative Unities is one of the main methods to reduce biodiversity loss in its extension. The project described here aims to compare the environmental conditions of streams, utilizing ecophysiological parameters that can indicate the presence of pollutants. For that, abiotic data and fish of a single genera (Hemigrammus) were collected inside the Guaribas Biological Reserve (Paraíba, Brazil) and outside it, subjected to different human interference. From the collected specimens, physiological markers were used as uniformity parameters for the distinct relationships between the fish from the study, being the quantification of total plasmatic proteins, moisture content of tissues, and the activity of MXR phenotype. Although the last one had inconclusive results, all of the parameters statistically demonstrated the existence of differences between the two groups and a more pronounced uniformity between the sites within the reserve. Combining this with the fact that fish from outside the reserve had a higher concentration of plasmatic proteins and also higher amounts of ammonia in the water of these sites, it was possible to argue that the conclusions regarding the effectiveness of the Conservative Unity were highly positive.

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1 Introduction

The devastation of rivers and streams from the Atlantic forest, biogeographical province (Udvardy, 1975) included in the list of the world's Hotspots (high biodiversity rates, endemism and anthropic pressure - (Myers et al., 2000), is caused mainly by pollution, deforestation, damming and introduction of exotic species. Pollution decreases abundance and richness of loca0l fish, being aggravated by loss of riparian vegetation when there is deforestation or change of river course due to damming. These population reductions are also caused in the presence of invasive species, introduced accidentally or intentionally by humans, modifying the original competition/prey relation (Miranda, 2012). With about 2-5% of the original vegetation preserved in Atlantic forest, one of the main strategies for conserving and preserving biodiversity from this and other biogeographical provinces is the delimitation of areas known as Conservative Units (CU), regulated in Brazil by the National System of Conservative Unit (SNUC) and implemented by Chico Mendes Institute (ICMBio) to be managed at federal, state and local levels (Arruda et al., 2013; Miranda, 2012).

Ecological data, such as richness and species abundance, are the most studied and represent how aquatic animals deal with general pollution, but the toxicity responsible for behavioral change depends on the organism, time of exposure and particle type, which demonstrates the need for physiological studies, specially in natural environments (Amoatey & Baawain, 2019). As physiological, morphological and behavioural aspects represent how their functions are performed in the ecosystem, they are nominated functional characteristics, commonly studied by parameters of strength and resilience. Changes on these parameters reflect environmental changes that could not be seen by parameters of taxonomical diversity, which happens when different species possess similar functions, highlighting how these, still scarce studies, can be promising (Júnior, 2021). Taking it into account, as also the relevance of Conservation Units for the Atlantic forest species, this study brings up the possibility to compare the ecological health from these waterbodies by identifying physiological stress markers from the fish within and near CU's.

One of the main physiological functions affected by pollutants (such as pesticides, heavy metals, drugs and microplastics) is the osmoregulation one, that, as well as the pH control, is predominantly done by gills. Being also critical to gas exchange, ionic regulation and nitrogen excretion, its functions are partially performed by hormonal control of blood flow (carried out by catecholamines, which reduce vascular resistance in the gills and increase blood flow in the efferent artery via beta and alpha-adrenoreceptors, in that order) and mainly by the transport of ions, probably made in the chloride cells from the filament epithelium, that possess transport enzymes called sodium/potassium-dependent ATPases (Evans, 1987).

The action of toxic compounds in the gills leads to a series of histopathologies in response to stress, such as necrosis and epithelial desquamation, frequently generated by flaws in cellular osmoregulation due to ATPase inhibition and physiological failure of the chloride cells or gills as a whole. The effects of this action on the organism can be measured through a few blood markers, mainly lower levels of Chlorine and Sodium ions, obtained by freshwater fish through exchange with internal ions (Evans, 1987), and higher levels of plasmatic proteins (Sabae & Mohamed, 2015). The hydration level of gills and muscles is another physiological parameter tied to the animal's capacity of dealing with salinity variation, and, for that, suffers mensurable effects from the action of toxic compounds in the environment. This active control parameter was suggested by Ayrapetyan (2012) as a universal biomarker for detecting pollution in the environment, being upheld by David et al. (2018) in a study with osmoconforming and euryhaline oysters under effect of different - but common - levels of salinity, and, more discretely, by Sabae & Mohamed (2015) in a study with freshwater fish.

Another relevant biomarker is the abnormal activity of the multixenobiotic resistance (MXR) phenotype, an innate gene of many aquatic species that is expressed in organs such as gills, kidneys, blood-brain barrier, liver and pancreas. It is activated by exposure to pollutants and results in the production of P-glycoproteins (Pgp) in the membrane, which act to remove endogenous or exogenous toxic compounds out of the cell, protecting DNA and inducing its excretion. Although the MXR activity explains how some species are more resistant to the presence of pollutants, some substances can reverse its effectiveness, being called chemosensitizers (or simply MXR-inhibitors). Those compounds can cancel the Pgp activity, alter the mechanism regulation, saturate or even reverse MXR activity, acting, in this case, by saturating the Pgp pump and allowing other xenobiotics to enter the cell and increase its toxicity, even if the chemosensitizers themselves are not toxic. Precisely because they are mostly organic, natural, primarily harmless and very present in the materials emitted anthropogenic in nature, its detection should also be a priority in environmental risk studies (Smital & Kurelec, 1998).

The hypothesis of markers linked to physiological stress of fish providing evidence of the environmental stress levels occurring inside and outside of a Conservation Unity was tested in the Guaribas Environmental Reserve (ReBio Guaribas), one of the 22 CU's inside the Atlantic forest area of Paraíba state, located in the Zona da Mata, wettest and most economically relevant mesoregion from the state (Fig. 1). Grouped, according to SNUC, as one of the integral protection areas (aimed to conserve biodiversity, being minority within the biogeographical province – 33,45% of the CU's), the definition of environmental reserves prohibits human interference inside their limits, except for management actions that aim its preservation and recovery. Despite being one of the most restricted types of Conservation Units, there are 22 populations that influence ReBio Guaribas due to historic factors of occupation and absence of an effective buffer zone around the three areas that make up the reserve. Those populations are mainly rural and cause direct impact, by using wood and plants from the reserve, and indirect, causing siltation of rivers and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil, promoting possible comparable ecophysiological responses in the ichthyofauna inside and around the CU (Arruda et al., 2013; Ministério do Meio Ambiente, n.d.).

The ecophysiological evaluation of human impact on the forest, which is close to areas of monoculture and natural vegetation modified for livestock, was mainly guided by the fish species inventory of ReBio Guaribas made by Gouveia et al. (2017), in a study that lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - Oreochromis niloticus, second most farmed fish species in the world and tolerant to osmotic variation, and Poecilia reticulata, largely used in mosquito control and resistant to stressful environments (Miranda, 2012) - and none are listed as threatened.

The predominance of small fish inhabiting small basins explains the higher level of endemism in streams, resulting from the allopatric speciation generated by the low dispersion made by these fish and the consequent isolation of populations (Gouveia et al., 2017). Having in mind that these stream fish are phylogenetic close, this paper offers representative data for the entire local ichthyofauna using species of Hemigrammus genus, of the Characiformes order, characterized by a black conspicuous spot delimited on the dorsal fin. Its type species, Hemigrammus unilineatus, has a compressed and slightly elongated body and exhibits wide geographical distribution both in Central and South America, with registers in other Brazilian states, such as Alagoas and Pernambuco (Serra, 2010).

The identification of physiological stress biomarkers in the collected fish aims to evaluate the ecological health of sites inside and surrounding the CU, through quantitative comparison of the influence of anthropogenic action with data from a legally protected area (statistical analysis made in the integrated programming environment R and R Studio - (R Core Team, 2017; RStudio Team, n.d.)). This promotes a more tangible insight in contrast with other scientific papers that only make qualitative assessments of Conservative Units, and also using more specific markers than articles that only explore the ecological consequences of pollution exposure.

2 Material and Methods

2.1 Collection and animal maintenance

The sampling sites (Fig. 2) were cataloged and numbered by the co-author of this paper (Prof. Dr. Elvio Sérgio Figueredo Medeiros) in previous works, using three data points inside the Reserva Biológica Guaribas (n=92 fish collected) and three in the surrounding area (n=97), numbered, in order, P5 (n=29), P6 (n=31) and P7 (n=32), and P8 (n=28), P9 (n=35) and P10 (n=34), between October and November 2023.

In order to uniform the data, only fish from Hemigrammus Gill 1858 (Eschmeyer et al., 2016) genera were analyzed, being found in every sampling site and in abundance, captured in the shallow waters of streams using a hand net in sweeping motion. To facilitate handling, the specimens were transported in plastic bags with air and local water to the Animal Ecophysiology Laboratory (LEFA) from UEPB - campus V (João Pessoa, Brazil). To analyze the osmoregulatory function of the fish, blood samples were collected from some specimens (38 fish from inside the reserve and 40 from outside, a total of 78). For that, those specimens were anesthetized with eugenol, and a blood aliquot was obtained by puncturing the heart using a pipette with a heparinized tip. The samples were kept on ice, and plasma was subsequently separated from cells by centrifugation. Due to low volume of obtained plasma, because of the small size of fish, the chlorine ions assay could not be performed. From those same specimens, gill and muscle were collected in order to calculate the moisture content of tissues. Plasma and tissues samples were stored in a freezer at -20°C until the analyses were performed. The other part of collected specimens (54 fish from inside the reserve and 57 from outside, a total of 111) was used in the multixenobiotic resistance (MXR) phenotype assay.

Abiotic data, such as water temperature, dissolved oxygen, pH and multiple nutrients were simultaneously collected with the fish, allowing additional statistical analysis that may point out possible correlations between physiological biotic data and these environment components (analysis performed in the integrated programming environment R and R Studio - (R Core Team, 2017; RStudio Team, n.d.)).

2.2 Plasmatic protein dosage

The plasma samples were defrosted and an aliquot of 2 µL of each was diluted with 8 µL of distilled water (proportion) to perform the total protein dosage by the Bradford (1976) method, in which the dye was also diluted in   proportion. The intensity of each sample coloration was measured with a microplate reader (SpectraMax i3) at 595 nm wave length, and compared to a standard curve (BSA protein). The results were transferred to a spreadsheet, where the average value in milligrams of protein dosage was calculated for each sampling site, and a Kruskal-Wallis test was performed to determine the existence of differences between the averages of the points inside and outside the Reserva Biológica Guaribas, calculated in the R Studio program (RStudio Team, n.d.). Next, an analysis of variance (ANOVA) was performed to determine where there were differences between each sampling site.

2.3 Moisture content

The tissues samples (gill and muscle) were individually stored in 2 mL eppendorf tubes (previously weighted), initially weighted with the tissue moist, and, after 24 hours inside a 60°C oven, weighted with the dry tissue. The results were transferred to a spreadsheet, where it was calculated the percentual content of tissue water content and the average valor for each sampling site. A T-test (gill) or Krustal-Wallis test (muscle) was performed to determine if there were significant differences between the average values inside and outside the Conservative Unity, followed by an ANOVA to determine the difference between all six sampling sites.

2.4 MXR phenotype activity

The multixenobiotic resistance (MXR) gene activity was analysed through the rhodamine B accumulation assay adapted from Smital & Kurelec (1998). Rhodamine B is a substrate of P-glycoprotein, the molecular basis of the MXR phenotype. Thus, after exposing the fish to this substrate, the analysis of the amount of rhodamine accumulated in the animal's tissues (mainly the gills) reflects the activity of the MXR phenotype: the greater the accumulation, the lower the activity, and the lower the accumulation, the greater the activity. Therefore, specimens of fish from each sampling site (three inside the Conservative Unity and three outside, total of six sampling sites) were transferred to plastic aquariums containing dechlorinated water and rhodamine B at 2,5 µM concentration, staying in this condition for one hour. The aquarium remained under constant aeration and protected from direct light. After exposition, the fish were anesthetized with eugenol and sacrificed for gill sampling. The tissue samples were allocated in Eppendorfs tubes and weighted in an analytical scale. The weight of moist tissue was obtained by subtracting the weight of the previously weighted empty eppendorf tubes.The tissues were then homogenized with 500 µL of distilled water and the supernatant was transferred in triplicate of 100 µL to a 96-well microplate. The fluorescence intensity of the supernatant (corresponding to the intracellular rhodamine B fluorescence in the tissue) was measured in the microplate reader (SectraMax i3) at 544 nm excitation and 590 nm emission. The fluorescence value was then normalized by the moist tissue weight in milligrammes, and a Kruskal-Wallis test was calculated in R Studio (RStudio Team, n.d.) to determine if there were differences between the averages inside and outside the Conservative Unity, followed by an ANOVA to visualize the differences between each sampling site.

3 Results

3.1 Plasmatic protein dosage

The average plasmatic protein concentration, parameter that ispositively correlated to environmental stress, was of 32,2 mg in the sampling sites within the Reserva Biológica Guaribas (n = 18 plasma samples) and 80,1 mg in the outside sites (n=26), revealing an almost 150% higher concentration in the sites with greater human influence. The total number of plasmatic protein dosages (n=44) was smaller than the total number of collected fish for this part of the analysis (n=78) because of the difficulty on collecting blood from smaller specimens, in addition to the need to collect a minimum amount of plasma required by the Bradford method. After checking the non-normality of variance of the total protein data, a non parametric Kruskal-Wallis test confirmed the existence of statistically significant difference (p-value<0,005) between the data collected from fish inside and outside the Conservative Unity (Fig. 3).

```r
library(readxl)
library(gplots)
library(agricolae)
library(sciplot)
library(PMCMRplus)
library(corrplot)
library(psych)
library(scales)
library(visreg)
citation()
```

```
dir()
setwd("C:/Users/ellen/Downloads/Manuscritos/mxr23_Q")
tcc=read.csv("Dados_TCC.csv", h=T, stringsAsFactors = T, dec=",")
#TP
hist(tcc$TP, probability = T)
lines(x=density(x=tcc$TP, na.rm=T, adjust=2)) #parece um pouco
shapiro.test(tcc$TP) #0,001 = não normal
bartlett.test(tcc$TP~tcc$Local) #0,000001765 = não homo ###kruskal
kruskal.test(tcc$TP~tcc$Local) #0,00000002328 = diferem

par(mfrow=c(1,1),mar=c(3,5,1,1),lwd=1)
boxplot(tcc$TP~tcc$Local)
boxplot(tcc$TP~tcc$Local,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgree
box(bty="o", lwd=2)
axis(side=2,las=1,font=2)
nomes=c("INSIDE","OUTSIDE")
pos=c(1,2)
axis(side = 1,at = pos, labels = nomes, las=1,font=9)
mtext(text="Total Plasmatic Proteins (mg)", side=2,line=3,cex=1,font=7)
legend("topleft", legend=c("Kruskal-Wallis test demonstrated", "statistical difference", "(p-
```

The subsequent ANOVA test confirmed the uniformity of values from the sites inside the reserve (all grouped as “a”), differently from the outside sites (Fig. 4).

```
anova(lm(formula=tcc$TP~tcc$Site)) #0,00000000000002783 = diferença
(tkPT=HSD.test(y=lm(formula=tcc$TP~tcc$Site),trt="tcc$Site",group=T))
boxplot(tcc$TP~tcc$Site) #a,a,a,c,b,bc #dentro e fora muito mais isolados

par(mfrow=c(1,1),mar=c(3,5,1,1),lwd=0.75)
boxplot(tcc$TP~tcc$Site,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgreen
box(bty="o", lwd=2)
axis(side=2,las=1,font=2)
nomesP=c("P5","P6","P7","P8","P9","P10")
posP=c(1,2,3,4,5,6)
axis(side = 1,at = posP, labels = nomesP, las=1,font=9)
mtext(text="Total Plasmatic Protein (mg)", side=2,line=3,cex=1,font=7)
tapply(tcc$TP,tcc$Site,max, na.rm=T)
text(x=c(1,2,3,4,5,6),y=c(34,35.2,35.6,98,98,87.7)+7,labels=c("a","a","a","c","b","bc"),cex=
```

The increase of total plasmatic protein concentration in the animals exposed to pollution is consistent with the (Sabae & Mohamed, 2015) study, which attributes this increase to five possibilities: activation of metabolic systems as answer to pollutants exposure, degradation of cellular material in the liver, severe pathological conditions in the liver and kidneys, loss of water in the plasma, and induction of proteic synthesis in the liver.

3.2 Moisture content

The average values of moisture content, parameter related to the osmoregulatory function, were 84,3% for the gill and 80,3% for the muscle of fish inside the ReBio Guaribas (n=38 fish). Regarding the fish from sampling sites surrounding the reserve (n=40), the averages were 88,9% for the gills and 82,3% for the muscle, increasing, in order, only about 5,6% and 2,5% the values from inside, results that are not very expressive, like the ones described by Haredi (2020). However, through T test (gills, which data had normal variation) and Kruskal-Wallis test (muscle, with non-normal variation data), it was detected statistical difference for both tissues between fish from the sampling sites inside and outside the Guaribas (Fig. 5).

```r
#TH_B = GMC
hist(tcc$GMC, probability = T)
lines(x=density(x=tcc$GMC, na.rm=T, adjust=2)) #quase
shapiro.test(tcc$GMC) #0,06 = normal
bartlett.test(tcc$GMC~tcc$Local) #0,7455 = homo ###teste t
t.test(tcc$GMC~tcc$Local) #0,002728 = diferem

par(mfrow=c(2,1),mar=c(2,5,1,1),lwd=1)
boxplot(tcc$GMC~tcc$Local)
boxplot(tcc$GMC~tcc$Local,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgre
box(bty="o", lwd=2)
axis(side=2,las=1,font=2)
nomes=c("INSIDE","OUTSIDE")
pos=c(1,2)
axis(side = 1,at = pos, labels = nomes, las=1,font=9)
mtext(text="Gill Moisture", side=2,line=3.5,cex=0.9,font=7)
mtext(text="Content (%)", side=2,line=2.7,cex=0.9,font=7)
legend("top", legend=c("p-value=0,002728"), text.font = 2, bty="n", cex=0.55, text.col="gray2

#MMC
hist(tcc$MMC, probability = T)
lines(x=density(x=tcc$MMC, na.rm=T, adjust=2)) #parece
shapiro.test(tcc$MMC) #0,002 = não normal
bartlett.test(tcc$MMC~tcc$Local) #0,047 = não homo ###kruskal
kruskal.test(tcc$MMC~tcc$Local) #0,000004813 = diferem

par(mfrow=c(1,1),mar=c(1,5,5,1),lwd=1)
boxplot(tcc$MMC~tcc$Local)
boxplot(tcc$MMC~tcc$Local,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgre
box(bty="o", lwd=2)
axis(side=2,las=1,font=2)
nomes=c("INSIDE","OUTSIDE")
```

```
pos=c(1,2)
axis(side = 1,at = pos, labels= nomes, las=1,font=9)
mtext(text="Muscle Moisture", side=2,line=3.5,cex=0.9,font=7)
mtext(text="Content (%)", side=2,line=2.7,cex=0.9,font=7)
legend("top", legend=c("p-value=0,000004813"), text.font = 2, bty="n", cex=0.55, text.col="g
```

The ANOVA test demonstrated similarity between data from sites inside ReBio Guaribas (P5, P6 and P7) and P8, and a lesser similarity with P9, that was also close to P10 (Fig. 6).

```
anova(lm(formula=tcc$GMC~tcc$Site)) #0,0000004971 = diferença
(tkTHB=HSD.test(y=lm(formula=tcc$GMC~tcc$Site),trt="tcc$Site",group=T))
boxplot(tcc$GMC~tcc$Site) #a,a,ab,a,bc,c ##P8 próximo aos de dentro, p9 e p10 mais isolados

par(mfrow=c(1,1),mar=c(2,5,1,1),lwd=0.75)
boxplot(tcc$GMC~tcc$Site,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgree
box(bty="o", lwd=2)
axis(side=2,las=1,font=2)
nomesP=c("P5","P6","P7","P8","P9","P10")
posP=c(1,2,3,4,5,6)
axis(side = 1,at = posP, labels = nomesP, las=1,font=9)
mtext(text="Gill Moisture", side=2,line=3.5,cex=0.9,font=7)
mtext(text="Content (%)", side=2,line=2.7,cex=0.9,font=7)
tapply(tcc$GMC,tcc$Site,max, na.rm=T)
text(x=c(1,2,3,4,5,6),y=c(91.3,94.3,98.1,89.1,97.1,97.9)+3,labels=c("a","a","ab","a","bc","c
```

The overall statistical analysis performed for moisture content reinforces this as a more discreet and inefficient parameter to make direct quantitative comparison for animals exposed to low salinity variation from the environment, but still accurate in determining the uniformity degree of more representative groups (i.e., fish from inside and outside the Conservative Unity).

3.3 MXR phenotype activity

The average values of normalized fluorescence measured in fish gills were 25.449 rfu/mg (relative fluorescence units/milligramme of tissue) for inside the Conservative Unity (n=54 fish) and 67.138,3 rfu/mg for outside (n=57), value nearly 2,64 times bigger. This difference between averages is opposite to the expected (higher fluorescence signals a lower MXR phenotype activity, meaning a lower exposition to pollutants), that can be explained by the presence of MXR phenotype inhibitors, which were highly correlated to more polluted waters by Kurelec et al. (2000). Among the possible pollutants presents, pesticides, fragrances, microbial degradation products and natural inhibitors from invasive species were described as compounds with high affinity to the Pgp or inhibiting the PKC (protein kinase C) modulator, resulting in high accumulation of rhodamine B (Smital, 2004) Despite the unexpected result, the variance difference between data from the MXR phenotype activity inside and outside the ReBio Guaribas (confirmed by the Kruskal-Wallis test and illustrated in Fig. 7)

```
#MXR
hist(tcc$MXR, probability = T)
lines(x=density(x=tcc$MXR, na.rm=T, adjust=2)) #não parece
shapiro.test(tcc$MXR) #0,0000038 = não normal
bartlett.test(tcc$MXR~tcc$Local) #0,000264 = não homo ##kruskal
kruskal.test(tcc$MXR~tcc$Local) #0,000000000005607 = diferem

par(mfrow=c(1,1),mar=c(3,6,1,1),lwd=1)
boxplot(tcc$MXR~tcc$Local)
boxplot(tcc$MXR~tcc$Local,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgre
box(bty="o", lwd=2)
axis(side=2,las=1,font=2)
nomes=c("INSIDE","OUTSIDE")
pos=c(1,2)
axis(side = 1,at = pos, labels = nomes, las=1,font=9)
mtext(text="Fluorescence/Mass of Tissue (rfu/mg)", side=2,line=4.5,cex=1,font=7)
legend("topleft", legend=c("Kruskal-Wallis test demonstrated", "statistical difference", "(p
```

was also the most conclusive when analysed by ANOVA test, being the only physiological parameter that shows total uniformity for both relationship with the reserve (Fig. 8).

```
anova(lm(formula=tcc$MXR~tcc$Site)) #0,0000000000005 = diferença
(tkMXR=HSD.test(y=lm(formula=tcc$MXR~tcc$Site),trt="tcc$Site",group=T))
boxplot(tcc$MXR~tcc$Site) #a,a,a,b,b,b #dentro e fora 100% isolados

dunnMXR=kwAllPairsDunnTest(formula=tcc$MXR~tcc$Site)
summary(dunnMXR) #ad,b,ab,c,c,cd

par(mfrow=c(1,1),mar=c(3,6,1,1),lwd=0.75)
boxplot(tcc$MXR~tcc$Site,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgree
box(bty="o", lwd=2)
axis(side=2,las=1,font=2)
nomesP=c("P5","P6","P7","P8","P9","P10")
posP=c(1,2,3,4,5,6)
axis(side = 1,at = posP, labels = nomesP, las=1,font=9)
mtext(text="Fluorescence/Mass of Tissue (rfu/mg)", side=2,line=4.5,cex=1,font=7)
tapply(tcc$MXR,tcc$Site,max, na.rm=T)
text(x=c(1,2,3,4,5,6),y=c(68400,31371,44811,88000,150658,121050)+10000,labels=c("a","a","a","
```

3.4 Abiotic data

The abiotic data from every sampling site water was measured and reunited in Tab. 1. The abnormal proportion of ammonia indicates the presence of effluents in the water outside the Reserva Biológica Guaribas, and, as discussed by (Piedras et al., 2006), this elevated concentration generates metabolic ammonia retention, causing toxicity. The same paper also points out that the toxicity of non-ionized ammonia in the aquatic environment is greater for smaller organisms, but depends on the interaction with other environmental variables, such as temperature, pH and salinity, demanding specific studies with the target species to determine their tolerance to ammonia.The Pearson correlation analysis among biotic data showed only one significant correlation, between Total Protein and MXR phenotype activity (0,6 - moderate and positive), and among abiotic data only two significance, between Nitrate e Phosphor (0,69 - moderate and positive) and between Dissolved Oxygen and Ammonia (-0,82 - strong and negative). It was also detected significant correlation among both data, between Gill Moisture Content and Ammonia (0,55 - moderate and positive), between Gill Moisture Content and Dissolved Oxygen (-0,62 - moderate and negative), and between MXR activity and Nitrate (-0,62 - moderate and negative).

4 Discussion

All parameters were effective to point out statistical differences between data from inside the ReBio Guaribas and its surroundings, highlighting - alongside the higher levels of ammonia in the water and plasmatic proteins in the fish - that changes in the environment are impacting the fish from the streams studied. Beyond that, the statistical analysis of physiological parameters from fish collected inside the Conservation Unity showed uniform results among its three sampling sites, consistent with what is expected from natural environments with little human influence. The same did not occur among the three sites outside the CU (with the exception of the MXR phenotype activity), indicating the action of different levels of environmental stress. The differences between the average results from the physiological parameters measured inside and outside the Conservation Unity were illustrated in the Fig. 9, highlighting the total protein as one that proved itself as an excellent biomarker to detect pollutants in the aquatic environment.

These preliminary obtained data were precise to indicate significant differences among the conditions in which the fish inhabited. But as much MXR phenotype activity data were as disparate as the ones of total protein and even more precise to point out the uniformity among sampling sites inside and outside the ReBio Guaribas, it showed values opposite to what was expected, with higher fluorescence in the tissue of fish outside the reserve, normally indicating lower MXR phenotype activity. This inconsistency can be explained by the possible presence of chemosensitizers in the water, since the fish did not went through a decontamination period, and thus, pollutants could still be bound to the active Pgp site and preventing rhodamine B to attach to it. The presence of such chemosensitizers can be quantified through the creation of a calibration curve, as described by (Kurelec et al., 2000), but, until done, the relevance of the MXR phenotype analysis must be restricted to the statistical context.

From the final results, an important positive evaluation can be made regarding the environmental protection that the Conservative Unity does within its extension, even when analysing one with historical problems of delimitation and still not entirely free of anthropic impacts. Adding the fact that it is located within the Atlantic forest, classified as a Hotspot, the verification of the efficiency of a CU in this biogeographical province is even more relevant, reinforcing the need for governments to create more legally protected areas as a key strategy to reduce global ecological crisis. This study also made an important description on the use of physiological biomarkers in ecological analysis, raising ecophysiology potential as a basis to future assays, which can be incorporated into effectiveness index such as seen in (Masullo et al., 2020) paper and presenting quantifiable data that can complement studies with analysis of historical and geographical data, such as the one from (Assis et al., 2021).

Conclusions

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Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Author1: Conceptualization, Data curation, Formal analysis, Investigation, Software, Visualization, Writing – review & editing, Writing – original draft. Author2: Data curation, Formal analysis, Software, Writing – review & editing. Author3: Data curation, Formal analysis, Software, Writing – review & editing. Author4:Data curation, Formal analysis, Software, Writing – review & editing. Author5: Conceptualization, Data curation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing. Author6: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing.

Ethical approval

This study did not require ethical approval as it did not involve human participants or sensitive data. Field surveys were conducted based on collection permit number (CEUA-048/2023 and SISBIO-89415-2).

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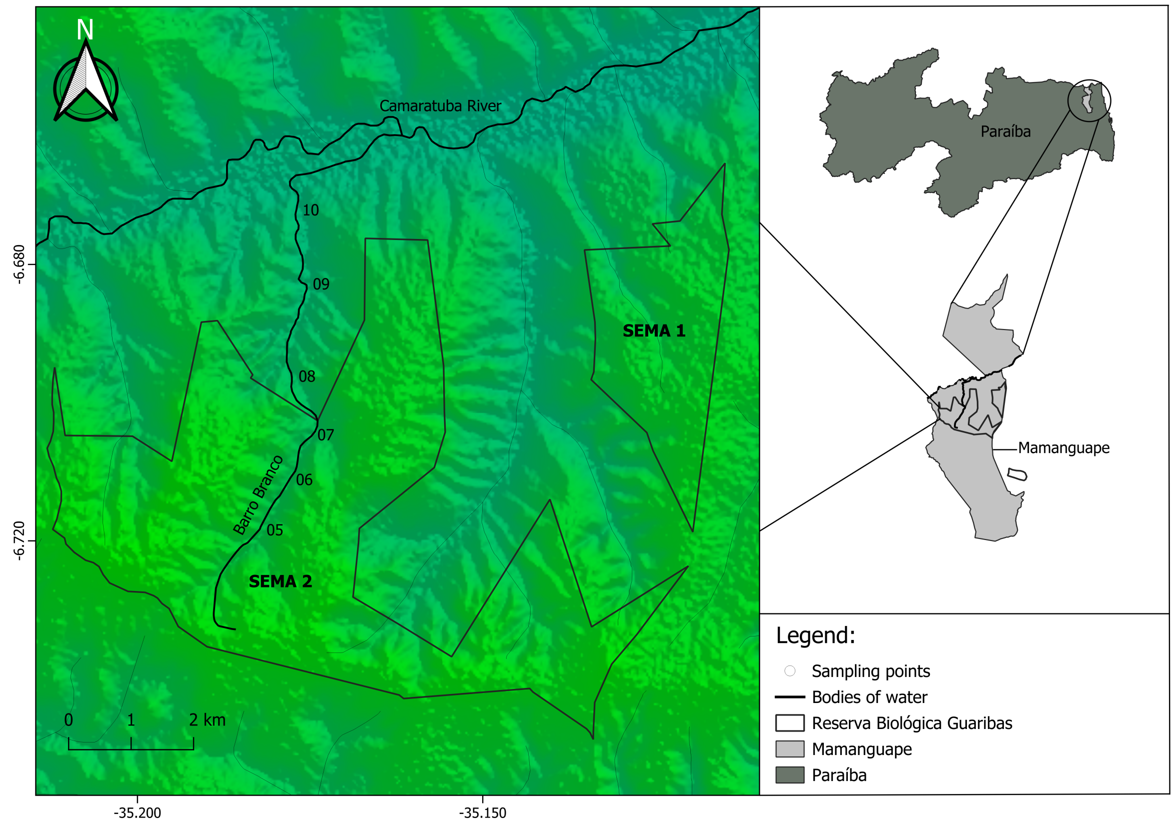
Figures and Tables

Figures



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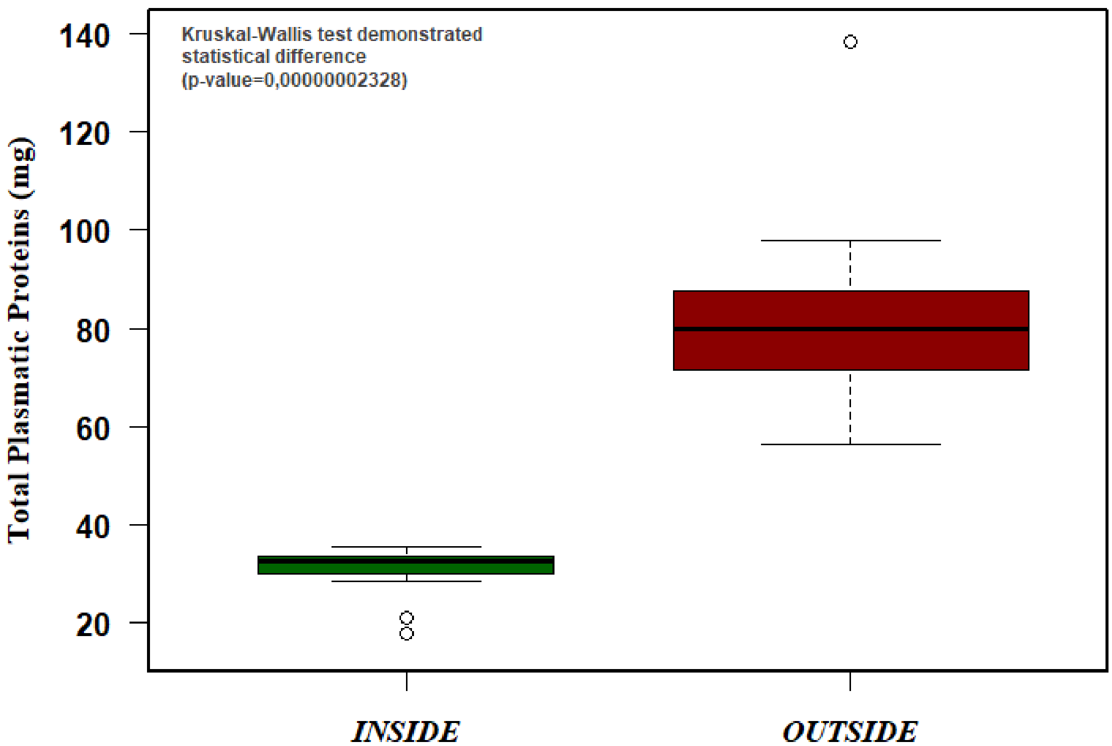

{#fig-Figure 3}

Tables

Apendices

Non-used figures and tables

> ⚠ Non-used codes